

## VENTSPILS, Latvia

WMO: 263140

Lat: 57.3956N

Lon: 21.5372E

Elev: 3

StdP: 101.29

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -15.2      | -11.5 | -18.1                         | 0.8 | -14.2 | -14.9 | 1.0 | -10.6 | 13.2                  | 2.1  | 11.5 | 2.0  | 2.6                   | 120  | 0.627 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.5                    | 26.8            | 19.5 | 24.2 | 18.4 | 22.6 | 17.8 | 20.7                | 24.7 | 19.5 | 23.1 | 18.5 | 21.7 | 3.9                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.2                            | 13.9 | 22.5 | 18.1 | 13.0 | 21.3 | 17.1 | 12.2 | 20.3 | 59.5          | 24.8 | 55.5 | 22.9 | 52.5 | 21.7 | 25.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 11.4          | 10.0 | 8.8 | DB                         | -17.9 | 29.6               | 4.9 | 3.1                                                | -21.4 | 31.8       | -24.3 | 33.6       | -27.0 | 35.3       | -30.6 | 37.6 | (4) |
| (5)               |      |     | WB                         | -18.2 | 22.2               | 4.8 | 1.8                                                | -21.7 | 23.6       | -24.5 | 24.6       | -27.2 | 25.6       | -30.7 | 27.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 7.5         | -1.2  | -1.6  | 1.3   | 5.8   | 10.2  | 14.0  | 17.1  | 17.0  | 13.2  | 8.4   | 4.2   | 1.2   |  |  |
|                                                                        | DBStd       | 7.51        | 4.88  | 5.06  | 3.25  | 3.62  | 3.88  | 2.98  | 2.76  | 2.53  | 2.73  | 3.24  | 3.63  | 4.45  |  |  |
|                                                                        | HDD10.0     | 1644        | 348   | 326   | 271   | 134   | 43    | 2     | 0     | 0     | 6     | 68    | 173   | 273   |  |  |
|                                                                        | HDD18.3     | 3991        | 606   | 559   | 529   | 375   | 255   | 134   | 58    | 57    | 156   | 308   | 423   | 531   |  |  |
|                                                                        | CDD10.0     | 739         | 0     | 0     | 0     | 10    | 48    | 123   | 221   | 217   | 101   | 19    | 1     | 0     |  |  |
|                                                                        | CDD18.3     | 44          | 0     | 0     | 0     | 0     | 2     | 6     | 20    | 16    | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 312         | 0     | 0     | 0     | 1     | 28    | 51    | 135   | 95    | 3     | 0     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 66          | 0     | 0     | 0     | 0     | 4     | 11    | 31    | 20    | 1     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 4.1         | 4.7   | 4.2   | 4.3   | 4.0   | 3.7   | 3.8   | 3.5   | 3.4   | 3.9   | 4.2   | 4.3   | 4.7   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 649         | 46    | 29    | 34    | 33    | 33    | 49    | 68    | 65    | 75    | 71    | 84    | 65    |  |  |
|                                                                        | PrecMax     | 1007        | 103   | 72    | 66    | 72    | 70    | 123   | 244   | 202   | 168   | 180   | 131   | 142   |  |  |
|                                                                        | PrecMin     | 432         | 6     | 6     | 0     | 3     | 5     | 1     | 6     | 0     | 15    | 15    | 19    | 0     |  |  |
|                                                                        | PrecStd     | 118         | 25    | 17    | 17    | 19    | 15    | 29    | 45    | 47    | 35    | 41    | 26    | 30    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 7.2   | 6.9   | 13.8  | 20.9  | 26.7  | 28.2  | 30.0  | 29.0  | 23.0  | 17.8  | 11.4  | 9.1   |  |  |
|                                                                        |             | MCWB        | 6.7   | 5.4   | 8.1   | 11.9  | 17.5  | 19.1  | 20.8  | 21.3  | 16.5  | 13.6  | 9.9   | 8.5   |  |  |
|                                                                        | 2%          | DB          | 6.1   | 5.2   | 8.7   | 16.9  | 22.9  | 24.2  | 26.9  | 25.9  | 20.8  | 15.2  | 10.6  | 8.1   |  |  |
|                                                                        |             | MCWB        | 5.5   | 4.3   | 5.9   | 11.2  | 16.2  | 17.8  | 20.1  | 19.4  | 16.3  | 13.1  | 9.4   | 7.3   |  |  |
|                                                                        | 5%          | DB          | 5.2   | 4.6   | 6.2   | 14.2  | 19.7  | 21.7  | 24.2  | 23.8  | 19.1  | 14.1  | 9.9   | 7.2   |  |  |
|                                                                        |             | MCWB        | 4.5   | 3.8   | 4.6   | 9.3   | 14.6  | 16.4  | 19.1  | 18.4  | 15.6  | 12.3  | 8.9   | 6.5   |  |  |
|                                                                        | 10%         | DB          | 4.2   | 3.9   | 5.0   | 11.9  | 16.8  | 19.4  | 22.3  | 22.0  | 17.9  | 13.1  | 9.0   | 6.5   |  |  |
|                                                                        |             | MCWB        | 3.5   | 3.1   | 3.8   | 8.1   | 12.8  | 15.5  | 18.5  | 17.9  | 15.1  | 11.6  | 8.2   | 5.7   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 6.7   | 5.6   | 8.5   | 13.4  | 18.7  | 20.7  | 22.4  | 22.3  | 18.1  | 14.5  | 10.6  | 8.6   |  |  |
|                                                                        |             | MCDB        | 6.8   | 6.2   | 12.1  | 18.2  | 24.4  | 25.8  | 26.9  | 26.1  | 21.4  | 16.4  | 11.0  | 8.7   |  |  |
|                                                                        | 2%          | WB          | 5.5   | 4.5   | 6.0   | 11.5  | 16.9  | 18.5  | 20.9  | 20.5  | 17.0  | 13.5  | 9.7   | 7.4   |  |  |
|                                                                        |             | MCDB        | 5.9   | 5.0   | 7.7   | 16.0  | 21.3  | 22.9  | 24.9  | 24.3  | 19.3  | 15.0  | 10.3  | 7.8   |  |  |
|                                                                        | 5%          | WB          | 4.5   | 3.8   | 4.8   | 9.9   | 14.9  | 17.1  | 19.8  | 19.3  | 16.2  | 12.7  | 8.9   | 6.6   |  |  |
|                                                                        |             | MCDB        | 5.0   | 4.3   | 5.8   | 13.6  | 19.3  | 20.6  | 23.5  | 22.4  | 18.4  | 14.0  | 9.5   | 7.1   |  |  |
|                                                                        | 10%         | WB          | 3.5   | 3.1   | 3.8   | 8.4   | 12.9  | 15.9  | 18.8  | 18.4  | 15.3  | 11.9  | 8.2   | 5.6   |  |  |
|                                                                        |             | MCDB        | 4.1   | 3.6   | 4.6   | 11.2  | 16.2  | 18.7  | 22.0  | 21.2  | 17.5  | 13.1  | 8.9   | 6.2   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 4.1   | 4.7   | 5.2   | 7.4   | 8.0   | 7.4   | 7.5   | 8.0   | 7.1   | 5.7   | 3.9   | 3.7   |  |  |
|                                                                        |             | MCDBR       | 3.2   | 3.6   | 7.6   | 12.2  | 13.6  | 12.7  | 12.2  | 12.1  | 10.5  | 7.6   | 4.1   | 3.1   |  |  |
|                                                                        | 5% WB       | MCWBR       | 3.2   | 3.2   | 5.4   | 7.1   | 7.4   | 7.2   | 6.6   | 6.5   | 6.9   | 5.8   | 3.9   | 3.1   |  |  |
|                                                                        |             | MCWBR       | 3.3   | 3.5   | 6.7   | 10.9  | 12.5  | 10.8  | 10.4  | 9.8   | 8.7   | 6.5   | 3.9   | 3.0   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.300 | 0.325 | 0.347 | 0.380 | 0.374 | 0.371 | 0.402 | 0.395 | 0.364 | 0.347 | 0.333 | 0.296 |  |  |
|                                                                        |             | taud        | 2.225 | 2.269 | 2.314 | 2.314 | 2.398 | 2.446 | 2.381 | 2.392 | 2.456 | 2.403 | 2.259 | 2.184 |  |  |
|                                                                        | Ebn at Noon |             | 554   | 710   | 802   | 831   | 862   | 869   | 830   | 809   | 783   | 684   | 505   | 438   |  |  |
|                                                                        |             | Edh at Noon | 51    | 75    | 93    | 109   | 107   | 104   | 109   | 101   | 82    | 66    | 50    | 40    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 0.33  | 0.94  | 2.40  | 4.26  | 5.79  | 6.17  | 5.73  | 4.54  | 2.97  | 1.36  | 0.43  | 0.22  |  |  |
|                                                                        | RadStd      |             | 0.06  | 0.14  | 0.28  | 0.48  | 0.47  | 0.29  | 0.50  | 0.37  | 0.34  | 0.17  | 0.08  | 0.03  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (28 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page